

Chemistry - Question 121

Original paper page 25; original question and option regions

121. Pick the correct statement

- (1) CO which is major pollutant resulting from the combustion of fuels in automobiles plays a major role in photo chemical drug
- (2) Photochemical smog occurs in day time whereas classical smog occurs in early morning
- (3) Classical smog has an oxidizing character while photochemical smog has reducing character
- (4) Classical smog is good for health but not photochemical smog

Ans : 2

Chemistry - Question 122

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122. Benzene on ozonolysis followed by reaction with $Zn + H_2O$ gives

(1) 3 moles of glycerol

(2) 3 moles of glyoxal

(3) 3 moles of glyoxalic acid

(4) 3 moles of acetylene

Ans : 2

Chemistry - Question 123

Original paper page 25; original question and option regions

123. Alkanes having odd carbons cannot be prepared in

A) Wurtz reaction

B) Frankland reaction

C) Kolbe's electrolysis

D) Sabatier – sendersen reaction

(1) B, C and D

(2) A, C and D

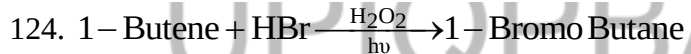
(3) A, B and D

(4) A, B and C

Ans : 4

Chemistry - Question 124

Original paper page 25; original question and option regions



The above reaction follows

(1) Markownikoff's rule

(2) Saytzeff's rule

(3) Anti Markownikoff's rule

(4) Hoffmann's rule

Ans : 3

Chemistry - Question 125

Original paper page 25; original question and option regions

125. Which one of the following has both schottky and Frenkel defects

(1) AgBr

(2) ZnO

(3) NaCl

(4) KCl

Ans : 1

Chemistry - Question 126

Original paper page 25; original question and option regions

126. For a dilute solution, Raoult's law states that :

- (1) The relative lowering of vapour pressure is equal to the mole fraction of solute
- (2) The relative lowering of vapour pressure is equal to the mole fraction of solvent

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Chemistry - Question 126 (continued)

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Ans : 1

Chemistry - Question 127

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127. Assertion (A) : 0.1M KCl , $0.1\text{M K}_2\text{SO}_4$ solutions cause the same osmotic pressure in solution

Reason (R) : Osmotic pressure depends on the number of particles present in the solution

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explanation of A.

(3) A is True but R is False.

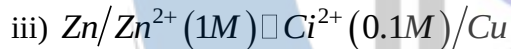
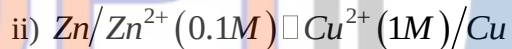
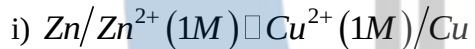
(4) R is True but A is False.

Ans : 4

Chemistry - Question 128

Original paper page 26; original question and option regions

128. The emf of the following three galvanic cells are represented by E_1 , E_2 and E_3 respectively. Which of the following is correct ?



(1) $E_1 > E_2 > E_3$

(2) $E_3 > E_2 > E_1$

(3) $E_3 > E_1 > E_2$

(4) $E_2 > E_1 > E_3$

Ans : 4

Chemistry - Question 129

Original paper page 26; original question and option regions

129. The rate of reaction $A \rightarrow \text{products}$ is 10 mole/lit/min at time $(t_1) = 2 \text{ min}$. What will be the rate in mole/lit/min at time $(t_2) = 12 \text{ min}$

(1) more than 10

(2) 10

(3) less than 10

(4) 20

Ans : 3

Chemistry - Question 130

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130. As the size of gold particle increases the colour of solution varies as

(1) Purple $\rightarrow$ blue $\rightarrow$ golden $\rightarrow$ red

(2) Golden $\rightarrow$ red $\rightarrow$ purple $\rightarrow$ blue

(3) Red $\rightarrow$ purple $\rightarrow$ blue $\rightarrow$ golden

(4) Blue $\rightarrow$ purple $\rightarrow$ golden $\rightarrow$ red

Ans : 3

Chemistry - Question 131

Original paper page 26; original question and option regions

131. Which of the following ore is used for the extraction of Zinc ?

(1) Bauxite

(2) Magnetite

(3) Malachite

(4) Sphalerite

Ans : 4

Chemistry - Question 132

Original paper page 26; original question and option regions

132. Nitrogen has unique ability to form $P\pi - P\pi$ multiple bonds with itself and with other elements due to

(1) Small size and low electronegativity

(2) Large size and high electronegativity

(3) Large size and low electronegativity

(4) Small size and high electronegativity

Ans : 4

Chemistry - Question 133

Original paper page 26; original question and option regions

133. Which of the following statement is incorrect ?

- (1) 4d, 5d, metals show higher enthalpies of atomization than those of 3d metals
- (2) d^5 configuration favours strong bonding between metal atom in d block
- (3) Gold has higher melting point than copper and silver

Chemistry - Question 134

Original paper page 27; original question and option regions

134. Assertion (A) : The difference in the successive oxidation states exhibited by transition elements is unity only

Reason (R) : d orbitals of transition elements are incompletely filled

(1) Both A and R are correct and R is the correct explanation of A.

(2) Both A and R are correct but R is not the correct explanation of A.

(3) A is True but R is False.

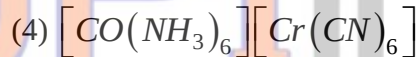
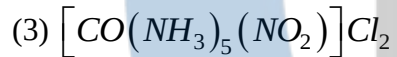
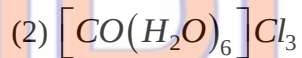
(4) R is True but A is False.

Ans : 1

Chemistry - Question 135

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135. Which of the following exhibits linkage isomerism ?



Ans : 3

Chemistry - Question 136

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136. Which chlorine oxide is used as bleaching agent for paper pulp and textiles ?



Ans : 1

Chemistry - Question 137

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137. Which of the following polymers has heterocyclic ring ?

(1) Bakelite

(2) Melamine

(3) Buna – S

(4) Terylene

Ans : 2

Chemistry - Question 138

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138. The bases that are common in both RNA and DNA are

(1) Adenine, Guanine, Thymine

(2) Adenine, Uracil, Cytosine

(3) Adenine, Guanine, Cytosine

(4) Guanine, Uracil, Thymine

Ans : 3

Chemistry - Question 139

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139. The antibiotic having β - Lactam ring

(1) Penicillin

(2) Chloramphenicol

(3) Streptomycin

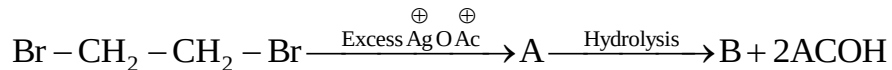
(4) Ciproflaxacin

Ans : 1

Chemistry - Question 140

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140. Identify A and B respectively in the following reaction



- (1) 1, 2 – Diacetoxy ethane and 1, 2 – Dibromo ethane
- (2) 1, 2 – Diacetoxy ethane and Ethylene glycol
- (3) Ethylene glycol and Glycerol
- (4) Ethylene glycol and 1, 2 – Diacetoxy ethane

Ans : 2

Chemistry - Question 141

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141. Predict the increasing order of acidity in the following compounds

Chemistry - Question 141 (continued)

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(3) Metanitrophenol < paranitrophenol < paracresol < phenol

(4) Paranitrophenol < metanitrophenol < paracresol < phenol

Ans : 2

Chemistry - Question 142

Original paper page 28; original question and option regions

142. Tertiary butyl methyl ether on treatment with HI forms

(1) Isobutane and Methyl iodide

(2) Isobutanol and Methanol

(3) Tertiary Butyl iodide and Methanol

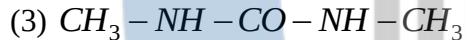
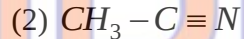
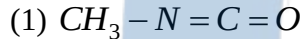
(4) Tertiary Butyl iodide and Methyl iodide

Ans : 3

Chemistry - Question 143

Original paper page 28; original question and option regions

143. The major gaseous product obtained in the following reaction is



Ans : 1

Chemistry - Question 144

Original paper page 28; original question and option regions

144. What are the total number of orbitals and electrons from $m = 0$, if there are 30 protons in an atom

(1) 6 orbitals, 12 electrons

(2) 5 orbitals, 10 electrons

(3) 7 orbitals, 14 electrons

(4) 4 orbitals, 8 electrons

Ans : 3

Chemistry - Question 145

Original paper page 28; original question and option regions

145. An iron ball has a mass of 35gms and a speed of 50 m/s . If the speed can be measured with an accuracy of 2% then the uncertainty in the position

- (1) $1.507 \times 10^{-34}\text{ m}$ (2) $1.507 \times 10^{-31}\text{ m}$ (3) $1.507 \times 10^{-33}\text{ m}$ (4) $1.507 \times 10^{-12}\text{ m}$

Ans : 3

Chemistry - Question 146

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146. In the periodic table an element with atomic number 56 belongs to

(1) IIIA group, 6th period

(2) IVA group, 5th period

(3) IIA group, 6th period

(4) IVA group, 6th period

Ans : 3

Chemistry - Question 147

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147. The first ionization potential of four consecutive elements present in the second period of the periodic table are 8.3, 11.3, 14.5 and 13.6ev respectively. Which one of the following is the first ionization potential of carbon

(1) 13.6

(2) 11.3

(3) 8.3

(4) 14.5

Ans : 2

Chemistry - Question 148

Original paper page 28; original question and option regions

148. CuCl has more covalent character than NaCl because

- (1) Na^+ has more polarizing power than Cu^+
- (2) Cu^+ has more polarizing power than Na^+
- (3) Cl^- has pseudo inert gas electron configuration

Chemistry - Question 149

Original paper page 29; original question and option regions

149. The bond orders in the pairs of bonded oxygen atoms in ozone molecule are

(1) $(1, 2)$

(2) $\left(\frac{1}{2}, 1\frac{1}{2}\right)$

(3) $\left(1\frac{1}{2}, 1\frac{1}{2}\right)$

(4) $\left(\frac{1}{2}, 2\frac{1}{2}\right)$

Ans : 3

Chemistry - Question 150

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150. The ratio of kinetic energies of 1g of H_2 and 2g of CH_4 at the same temperature will be

(1) 1 : 4

(2) 4 : 1

(3) 1 : 2

(4) 2 : 1

Ans : 2

Chemistry - Question 151

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151. At critical temperature surface tension of a liquid is

(1) Zero

(2) Infinite

(3) Varies liquid to liquid

(4) Cannot be measured

Ans : 1

Chemistry - Question 152

Original paper page 29; original question and option regions

152. 250ml of Na_2CO_3 solution contains 2.65g of Na_2CO_3 . 10ml of this solution is mixed with 'X' ml of water to obtain 0.001M Na_2CO_3 solution. The value of 'X' is

(1) 1000

(2) 990

(3) 9990

(4) 90

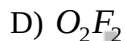
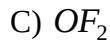
Ans : 2

Chemistry - Question 153

Original paper page 29; original question and option regions

153. Match the following

List – I



A B C D

(1) I II III IV

(3) III IV V II

List – II

I) Zero

II) + 1

III) – 1

IV) $-\frac{1}{2}$

V) + 2

A B C D

(2) II IV III V

(4) I IV IV II

Ans : 3

Chemistry - Question 154

Original paper page 29; original question and option regions

154. Which one of the following is correct for a spontaneous reaction at any temperature ?

(1) $\Delta H = -ve$, $\Delta S = +ve$, $\Delta G = -ve$

(2) $\Delta H = -ve$, $\Delta S = -ve$, $\Delta G = -ve$

(3) $\Delta H = +ve$, $\Delta S = +ve$, $\Delta G = -ve$

(4) $\Delta H = -ve$, $\Delta S = +ve$, $\Delta G = +ve$

Ans : 1

Chemistry - Question 155

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155. What is the ratio between the rate of decomposition of N_2O_5 to rate of formation of NO_2 ?

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Chemistry - Question 155 (continued)

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(1) 1 : 4

(2) 4 : 1

(3) 2 : 1

(4) 1 : 2

Ans : 4

Chemistry - Question 156

Original paper page 30; original question and option regions

156. Which one of the following statement is correct ?

- (1) The pH of $10^{-8} M$ HCl is 8.0
- (2) Bronsted – Lowry theory could not explain the acidic nature of BCl_3
- (3) The pH of 0.01M NaOH solution is 2.0
- (4) The K_w of pure water at 308K is $1.0 \times 10^{-14} \text{ mole}^2 / \text{lit}^2$

Ans : 2

Chemistry - Question 157

Original paper page 30; original question and option regions

157. AlH_3 is an example for

(1) Interstitial hydride

(3) Ionic hydride

(2) Electron rich hydride

(4) Electron deficient hydride

Ans : 4

Chemistry - Question 158

Original paper page 30; original question and option regions

158. Which one of the following is paramagnetic ?



Ans : 3

Chemistry - Question 159

Original paper page 30; original question and option regions

159. Which group P – block elements show highest positive oxidation state ?

(1) 16

(2) 17

(3) 18

(4) 15

Ans : 3

Chemistry - Question 160

Original paper page 30; original question and option regions

160. Even though carbon and silicon are non metals, they have higher melting points than others because

- (1) They exist as covalent solids in 3D networks
- (2) The bonds in their molecules are strong
- (3) They exhibit multiple bonding
- (4) They are highly electronegative

Ans : 1